

Lab: The Iteration Log — Tracking Learning Across Prototype Cycles

Objective

Build a systematic learning protocol for your invention project. You will practice documenting predictions, observing outcomes, analyzing prediction errors, classifying learning types (parameter vs. structure), and planning next iterations. By the end of this lab, you will have an iteration log template and at least one complete learning cycle documented.

Materials and Prerequisites

- Your invention idea and any prototype work from previous labs
- A notebook or digital document for your iteration log
- Optionally: a prototype to test during this lab (even a very simple one)
- No technical background required

Part 1: Establishing Your Baseline Model (10 minutes)

Before you can learn from experiments, you need to make your current generative model explicit. What do you currently believe about your invention?

State your current model of how your invention works (the mechanism):

Write your response here

List 5 key parameters in your model — specific values, proportions, or settings that you believe are important but are uncertain about:

#	Parameter	Current best guess	Confidence (1-10)
1			

Write your response here

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Write your response here

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Write your response here

|| 2 |

Write your response here

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Write your response here

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Write your response here

|| 3 |

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|| 4 |

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Write your response here

|| 5 |

Write your response here

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Write your response here

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Write your response here

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List 3 structural assumptions — causal relationships in your model that might be wrong entirely (not just the values, but the relationships themselves):

#	Structural Assumption	Why it might be wrong
1		

Write your response here

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Write your response here

|| 2 |

Write your response here

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Write your response here

|| 3 |

Write your response here

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Write your response here

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Part 2: Designing a Learning Experiment (10 minutes)

Choose one parameter or structural assumption to test. Design an experiment (physical prototype test, user observation, thought experiment with evidence gathering, or any other appropriate method).

What are you testing?

Write your response here

Is this a parameter test or a structure test? - ☐ Parameter test (adjusting a value within my current model) - ☐ Structure test (checking whether a causal relationship in my model is correct)

Specific prediction: "If I [action], then [expected outcome] because [reason from generative model]."

Write your response here

What would confirm this prediction? (Be specific and measurable)

Write your response here

What would disconfirm this prediction? (Be specific and measurable)

Write your response here

What would be ambiguous? (An outcome that does not clearly confirm or disconfirm — and what you would do about it)

Write your response here

Part 3: Running the Experiment (15 minutes)

If you can run the experiment now (even a simple one), do so. If not, imagine the experiment in detail based on your best knowledge, or describe the results of a past experiment.

Experiment Record

Date and time:

Write your response here

Exact action taken:

Write your response here

Exact observation (what actually happened — describe with precision):

Write your response here

Prediction error analysis:

Predicted	Observed	Error Magnitude	Error Direction

Write your response here

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Write your response here

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Write your response here

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Was the prediction error: - ☐ Small (model was approximately correct — parameter fine-tuning needed) - ☐ Large but interpretable (model was wrong in a specific way — clear update path) - ☐ Large and confusing (model was wrong but unclear how — more investigation needed) - ☐ Zero (prediction matched observation — little new learning)

Part 4: Model Updating (10 minutes)

Based on your prediction errors, update your generative model.

If Parameter Learning:

Which parameter needs adjustment?

Write your response here

Old value:

Write your response here

New value (based on evidence):

Write your response here

How confident are you in the new value? (1-10)

Write your response here

What experiment would further refine this parameter?

Write your response here

If Structure Learning:

Which causal relationship was wrong?

Write your response here

What is the new causal relationship?

Write your response here

What other parts of your model does this change affect?

Write your response here

How does this structural change affect your overall design approach?

Write your response here

Updated Model Summary

What has changed in your generative model since the start of this lab?

Write your response here

What new questions has this learning raised?

Write your response here

Part 5: Planning the Next Cycle (10 minutes)

Based on your updated model, plan your next experiment.

Next experiment target (parameter or structure):

Write your response here

Why this is the most important thing to test next:

Write your response here

Predicted outcome:

Write your response here

Design of the next prototype or experiment:

Write your response here

Map your learning trajectory:

Cycle	What was tested	Type (P/S)	Key learning	What changed
This lab				

Write your response here

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Write your response here

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Write your response here

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Write your response here

|| Next |

Write your response here

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Write your response here

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Write your response here

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Write your response here

|| After that |

Write your response here

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Write your response here

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Write your response here

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Write your response here

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At what point in your learning trajectory do you expect to shift from structure learning (exploring fundamentally different approaches) to parameter learning (optimizing within a chosen approach)?

Write your response here

Part 6: Building Your Iteration Log Template (5 minutes)

Create a reusable template for documenting future iterations. Fill in the headers:

Iteration Log Entry Template:

Field	Description
Date	When was this iteration done?
Iteration #	Sequential number
Hypothesis	What prediction is being tested?
Learning Type	Parameter or Structure?
Prototype Description	What was built/done?
Prediction	Expected outcome
Observation	Actual outcome
Prediction Error	The specific gap
Model Update	What changed in the generative model
Confidence Change	How did confidence change?
Next Action	What to do based on this learning

Commit to using this log for your next 3 invention iterations by writing a brief personal commitment:

Write your response here

Discussion and Debrief

1. **Learning classification:** Was your experiment's learning primarily parameter or structure? How did you make this determination? Was the distinction always clear?
2. **Surprise value:** How surprised were you by the results? In Active Inference terms, was the prediction error large enough to significantly update your model? If not, how could you design experiments that generate more informative surprises?
3. **Documentation discipline:** How difficult was it to make your predictions explicit before testing? Many inventors skip this step, but it is the foundation of systematic learning. What strategies will help you maintain this discipline?
4. **Cumulative learning:** If you have done multiple experiments (across this course or independently), do you see a learning trajectory? Is your uncertainty decreasing over time? Are you converging on a design or still exploring?
5. **Structure learning anxiety:** Structure learning can feel like failure — abandoning an approach you invested in. Have you experienced this? How can you recognize the difference between premature abandonment and necessary structure change?

Write your response here